

DEVELOPMENT AND IMPLEMENTATION OF THE Φ SAT-2 MISSION

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ABSTRACT

The Earth Observation directorate of the European Space Agency is working since 2020 at the design and implementation of the Φ -sat2 mission. The mission is based on the use of a compact 6U cubesat platform equipped with a combination of on-board processing capabilities, including AI, and a state-of-the-art multispectral instrument capable of medium to high-resolution imaging. The instrument, operates in the visible to near infrared (VIS/NIR) spectrum, acquiring eight bands (seven spectral bands along with a Panchromatic one).

Several AI applications have been selected to accompany the Φ sat-2 mission, each designed to address relevant challenges that can benefit from the deployment of edge-computing in space. 4 applications were selected since the beginning of the mission and developed in parallel to space and ground segment. These applications run on a dedicated framework and allow to retrieve usable streets from space in case of floods, provide vessel detection in coastal areas, identify and classify cloud in images and finally compress and reconstruct images on ground through the use of AI.

Alongside with the 4 main applications others have been selected through the OrbitalAI challenge ended in September 2023 and will be implemented before the integration of the spacecraft on the launch vehicle. To allow for a proper development of all the onboard applications a Φ sat-2 image simulator have been developed and it is able to generate synthetic data based on Sentinel-2 datasets. The Φ sat-2 spacecraft is currently under qualification, having reached a successful TRR in December 2023. The Qualification Acceptance Review (QAR) was held April 2024 while the launch on SpaceX Transporter 11 is expected at the beginning of Q2 2024.

After launch and LEOP the satellite will undergo a commissioning phase dedicated to the verification of its payload chain; after commissioning the mission will enter in its nominal phase that will last at least 12 months. During this period the applications will be routinely used and continuously tuned to finally meet their initial objectives.

1 INTRODUCTION

The Φ sat-2 mission is part of an European Space Agency (ESA) initiative to promote the development and implementation of EO innovative technologies combining the fast development and reduced cost of new space with innovative EO concepts. The first Φ sat related activities were kicked off in 2018 [1], with the aim of enhancing the already ongoing FSSCAT [2] mission with Artificial Intelligence (AI). Φ sat-1 experiment successfully flown in 2020 [3] demonstrating the benefits of edge computing through AI and paving the way for its successor: Φ sat-2.

Φ sat-2 is based on the results of the initial experiment indeed with the aim of designing a dedicated mission to fully explore the capabilities of having on-board processing.

In the following sections, a brief overview of the Φ sat-2 mission and its segments is given with in-depth insight on the on-board payload data chain building block. Subsequently, the on-board AI applications are described in order to conclude with the test plan, preliminary results and conclusions.

The spacecraft is equipped with a compact multispectral camera and a dedicated payload processing unit capable to handle multiple onboard applications, with an expected mission lifetime of at least 12 months after the commissioning phase (Figure 1). The following mission objectives were established at the beginning of the project:

- Demonstrate the enabling capabilities of running onboard Artificial Intelligence applications.
- Demonstrate relevance for applicative scenarios and operational missions.
- Demonstrate the ability of running multiple applications on board (either segregated or combined in a sequential way) and to update and upload them in different moment throughout the entire mission lifetime.

The consortium is led by Open Cosmos (UK) as Prime Contractor, CGI (IT) as coordinator of the various application providers and in charge of the development of NanosatMO Framework, Simera Sense Europe (BE) and Ubotica Technologies (IL) providing the payload main components (Multispectral Camera and AI accelerator) and KPLabs, CEiiA and GEO-K providing the remaining baselined applications.

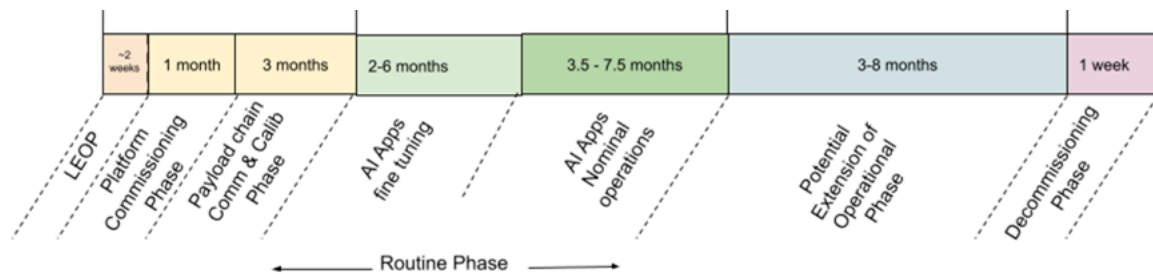


Figure 1: Mission timeline

In the context of the mission, the AI applications providers act as reference end-users for the final products. Moreover, the data produced by the mission will be freely available being distributed under an open and free data policy.

2 SPACE SEGMENT

The Φ sat-2 space segment is represented by the spacecraft, composed of the platform and the payload ensemble. The satellite will fly in a Sun Synchronous Low Earth Orbit at 510 km altitude; the design of the mission has been optimized to ensure the required Sun illumination conditions for both power generation and image acquisition. The Φ sat-2 platform is based on the standard Open

Cosmos Opensat in its 6U variant, as shown in Figure 2.

It is equipped with a state-of-the-art On-Board Computer (OBC) that includes a multicore processing unit, an ethernet based high-speed data switch, the mass memory, the GNSS receiver and the reconfiguration unit which is in charge of the system level FDIR.

On the main OBC is also running the enhanced version of the NanoSat MO Framework (NMF) [4], a software framework that facilitates the management, monitoring and control of the various onboard applications and their interaction with the platform.

The Attitude Determination Control System (ADCS) has been selected in order to meet mission performances and image quality requirements. It includes a processing unit, a series of magnetometers and Sun and Earth horizon sensors, a Star Tracker and a 3-Axis MEMS rate sensor. Four Reaction Wheels (RWs) are elastically mounted to the platform in a pyramidal configuration to maximize controllability and minimize the effect of the jitter on the payload. Magnetorquers are used for detumbling, sun-spin mode and desaturation of the RWs.

The payload is an ensemble of three different units, with distinct functions and connected through the platform main OBC as shown in Figure 3.

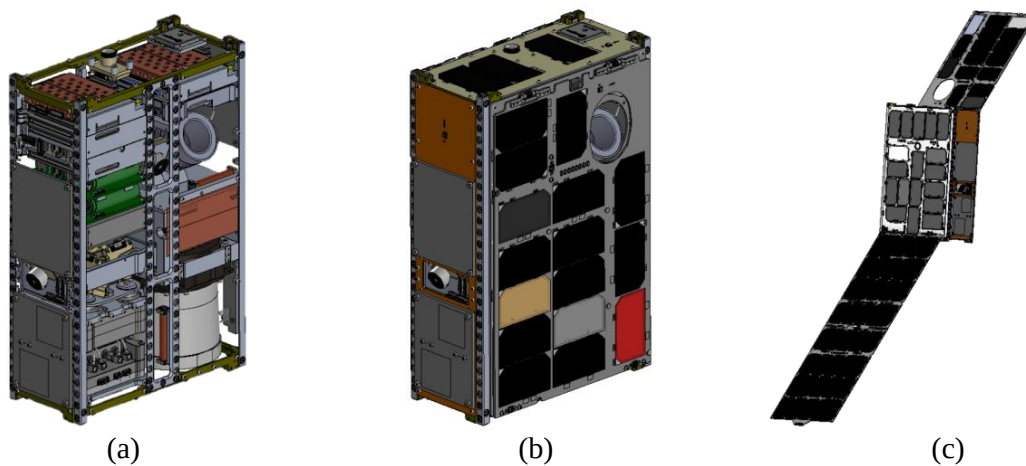


Figure 2: Φ sat-2 spacecraft internals (a), overall external layout stowed (b), deployed appendages (c)

The optical instrument is represented by SIMERA Sense Multiscope100 CIS (Figure 5), a push broom compact 1.5U multispectral camera. The payload is characterized by a GSD of 4.8 m and a swath of 19.7 km at 510 km altitude.

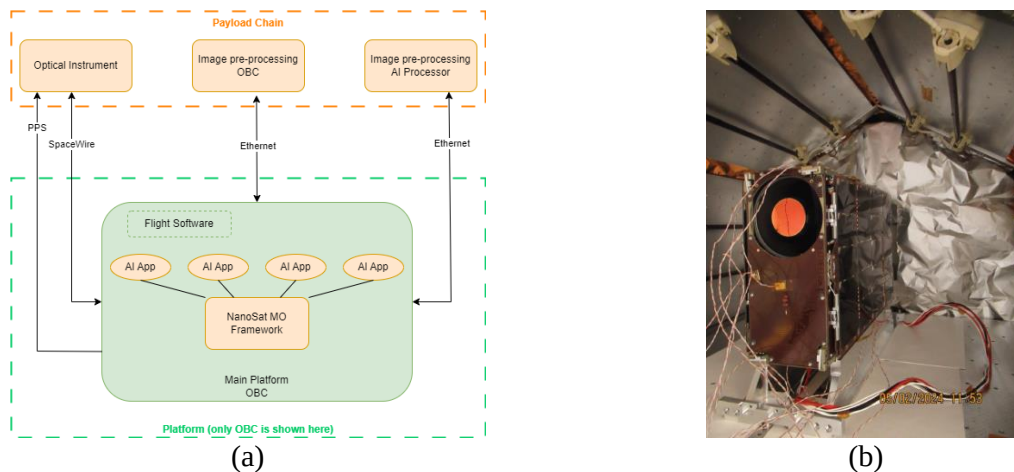


Figure 3: Φ sat-2 payload architecture (a), spacecraft nadir looking face during TVAC testing (b)

The on-board Digital Time Delay Integration (dTDI) operations are carried out automatically and

the number of usable stages depends on the spacecraft pointing stability. The image preprocessing OBC (i.e. secondary OBC) is a copy of the primary OBC, with the same processing capabilities and memory size with the sole exception of the GNSS receiver which is not present.

The third unit, completing the payload system, is represented by the AI processor. The Ubotica CogniSAT-XE1™ CubeSat Board serves both as a powerful edge computer and AI compute accelerator.

2.1 On-Board Processing Chain

To allow the AI applications to generate onboard data-products usable on ground without further processing, the Φsat-2 spacecraft has the capability to pre-process images on-board up to level 1C through a dedicated hardware-software system when required. Raw image files (Level 0) captured by the payload camera are transferred to the primary OBC through the spacewire link and then to the secondary computer that is hosting the Lev-0 to Lev-1C pre-processing library. The pre-processing steps are configured according to the specific requirements of each application and can provide different outputs as shown in Figure 4. The available functionalities at the time of launch will be: relative and absolute radiometric corrections, cosmetic filling, denoising, band co-registration and radiance to reflectance conversion. The library is also able to provide coarse onboard geolocation (200-400 m accuracy).

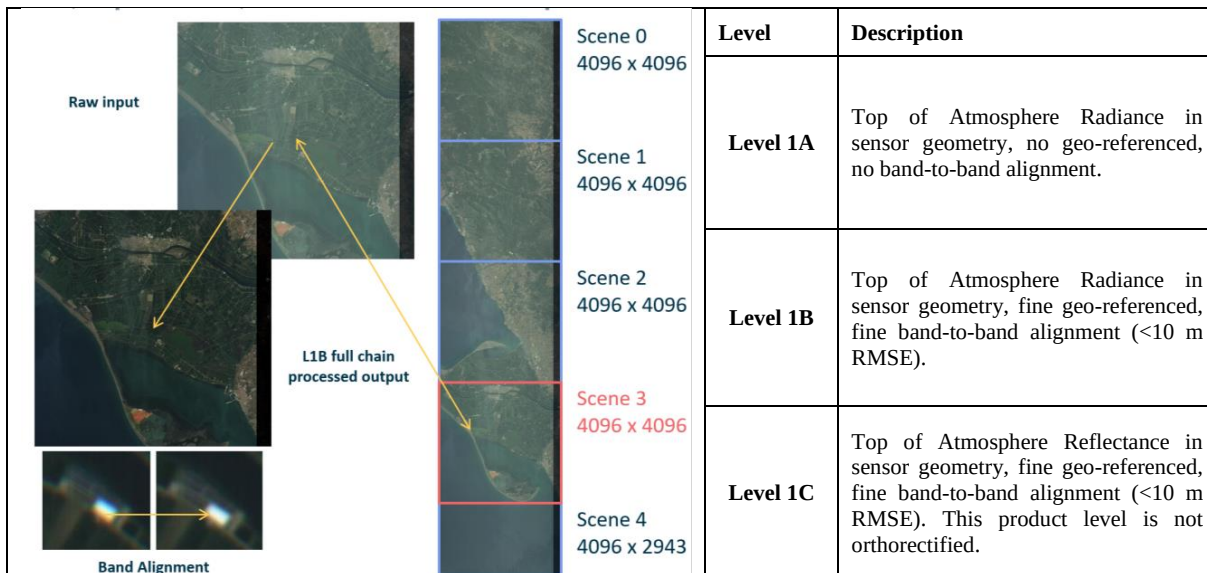


Figure 4: Data processing chain

Once the raw data has been pre-processed, all the relative files are then transferred back to the primary OBC and made available to the AI applications for their processing. Primary OBC is interfaced with the CogniSat AI processor that performs inference by running the AI models. The overall orchestration and interaction between the platform and the AI applications is done through the NanoSat MO Framework. Once the application provides an output (i.e. processed data), auxiliary data and metadata (e.g. GNSS, ADCS) can be added to the processed image to be finally downloaded through the X-Band transmitter.

The processing chain has been thoroughly tested during the implementation phase and the duration of the entire End to End chain has been monitored to check its compliance with the overall system requirements. The results of the test campaign conducted during the final integration of the spacecraft are provided in Table 1

Table 1: Φ sat-2 End to End performance results (on-ground campaign)

Square Image 4096x4096 pixels	Vessel Detection	Deep Compression v2.1.1	Deep Compression v2.2.1	Sat2MAP w/t Cloud Detection	Sat2MAP w Cloud Detection
Image Acquisition, capture, readout and pre-processing [s]	925.12	740.72	740.72	1425	1425
App Startup [s]	37.44	83.23	79.93	12	12
App Configuration [s]	0.06	5.06	5.06	34	34
App runtime [s]	120.61	209.83	347.54	216	217
App shutdown [s]	37.96	79.93	72.62	28	28
E2E Total [s]	1121.19	1118.77	1245.87	1715	1716
E2E Total [min]	18.7	18.6	20.8	28.6	28.6

The results show how the limit imposed by the power budget of a maximum of 30 min of continuous processing has been met by all the applications with ample margins in most of the cases. Nonetheless the obtained values are higher than what initially expected due to some limitations imposed in the implementation of the overall architecture. Further software updates during the course of the mission will likely improve the current situation.

2.2 GROUND SEGMENT

The core of the ground segment is represented by the Mission Operations Centre (MOC) under the responsibility of Open Cosmos, that will be using OpenApp (OPS module) to monitor the health of the satellite. OpenApp will be connected to the ground station providers network via two interfaces: an API, dedicated to scheduling bookings of the antenna, and a real time interface that directly transfers data to and from the antenna's baseband unit.

CGI will be using a Payload Operations Centre (POC) that enables easy tasking of the NanoSat MO Framework and its uploaded apps. For data transfer with the satellite, OpenApp will act as a relay, pushing along received data packets to the ground station baseband unit.

The ground station infrastructure will utilize the extensive KSAT ground station network. The mission baseline assumes that the ground station network in Svalbard is required to execute mission objectives.

3 AI APPLICATIONS

The baselined applications for the Φ sat-2 mission were proposed by the consortium during the selection phase of the and their development is expected to end with the final tuning phase once the mission has been launched.

3.1 SAT2MAP

Sat2Map is an AI software application developed by CGI, extracting street map data from satellite images. The idea behind the application was to be able to supply updated actionable information to the ground teams in emergency scenarios (e.g. a flooding event). In this specific case the end user on the ground would use the information provided by the application to support evacuations or the routing of rescuers.

The software implements Generative Adversarial Networks (GANs), a specialized neural network architecture capable of modelling the underlying data distribution in a given dataset such as to be able to create another data set following the same distribution. In case of Sat2Map, GANs are used to learn the data distribution of images to create images following the same distribution and create new images. In other words, GANs empower Sat2Map to learn how street maps derived from

satellite imagery shall look like and apply this knowledge to unknown satellite imagery. In general, GANs use two different networks, a Generator and a Discriminator network competing against each other. With the Sat2Map setting, the Generator creates new “fake” images mimicking the source images, i.e. learns to create images showing same data distribution as the input imagery. The Discriminator in turn learns to distinguish between fake and real imagery. The result of the Discriminator’s decision is subsequently fed back to the Generator, enabling the latter to adapt and improve the next set of fake images. Once the Discriminator is not able to distinguish real from fake imagery, the training concludes and the final model for generating target imagery is found. Conditional GANs are a slight variation of GANs that can further direct the results of a GAN into a desired direction. Within Sat2Map, the target direction is the generation of road maps based on satellite images.

Sat2Map uses three different implementations of such conditional GANs: CycleGAN, Pix2Pix and Jerin Paul. CycleGAN is a technique that involves the automatic training of image-to-image translation models without paired examples, whereas Pix2Pix uses paired training data. Lastly, Sat2Map uses an algorithm developed by CGI for model training. This algorithm is based on ideas presented by Jerin Paul. The models are trained using a collection of images from the source (i.e., satellite images) and target (i.e, street maps). These techniques are powerful, achieving visually impressive results on a range of diverse application domains. Finally, the Jerin Paul model has been chosen for the Sat2Map App for its final implementation. The model has been trained exploiting synthetic representative dataset generated by processing different mission including PlanetScope and S-2. The dataset was prepared aiming at enabling the model to learn distinguishing between streets, flooded streets, and cloudy areas (Figure 5).

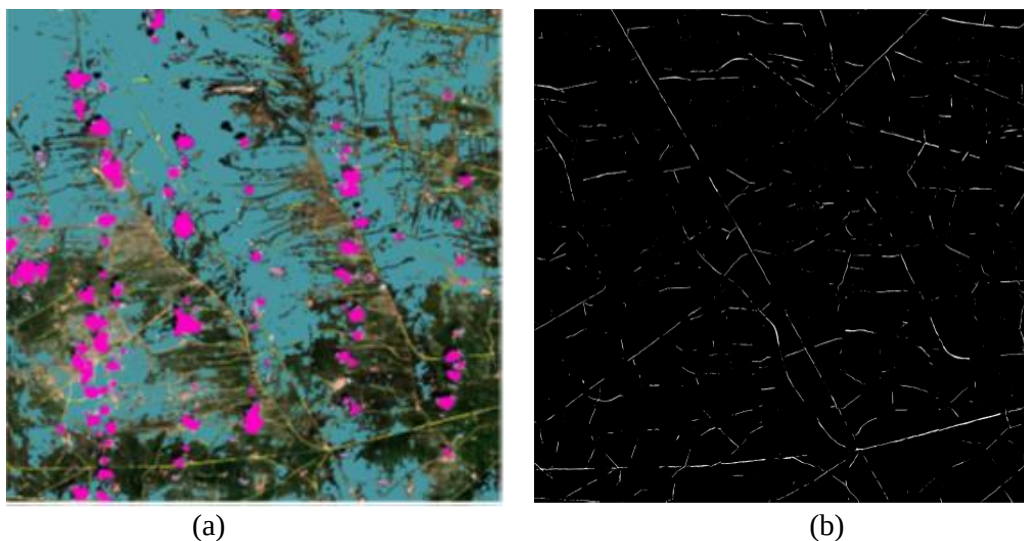


Figure 5: Example of street mapping output:

The model has been trained over a specific area in the Assam region (Northern India) for which relevant images including flooded streets were available and tested again over a different area with similar characteristics in Pakistan.

A series of metrics have been identified to evaluate the performance of the application from an end user perspective that will be further refined during the nominal phase of the mission. To support the on-ground users of the processed maps, a quick georeferencing procedure has also been designed to improve the quality of the provided information.

3.2 AUTONOMOUS VESSEL AWARENESS (AVA)

The Autonomous Vessel Awareness experiment (AVA) aims to demonstrate the capability to autonomously develop awareness about vessels in the maritime domain using AI. AVA will run

onboard Φ sat-2 and will make use of machine learning techniques (in particular deep learning) to autonomously detect and classify vessels in optical imagery captured by the Φ sat-2 camera (e.g., based on the following elements: absence of clouds, presence of ocean water, presence of ships in the image). It will then determine if a given scene (or areas) is interesting and requires further monitoring. Based on this, the sensor could further acquire data and transmit the data to the ground for further analysis. This will enable faster responses, and potentially higher value detections. AVA will enable:

- A reduction in the volume of downlinked images;
- A reduction in image processing and analysis time by human operators;
- A reduction in operational costs of satellite missions;

AVA results will be of interest to specific EO user communities such as maritime authorities and regulators in countries with coastal areas and Exclusive Economic Zones (EEZ), as well as their potential use in the detection of illegal fisheries and monitoring of fisheries in general.

The implementation of the model consists of two parts: image enhancement and vessel detection. For image enhancement, a deep learning model (Single Image Super-Resolution Model) is used. This model uses the concept of Convolutional Neural Network (CNN), intending to increase the original image quality, and consequently the performance of the whole AVA app (considering physical satellite limitations and onboard devices). Regarding vessel detection in sea and oceans satellite images, YOLO can be used both to detect and classify the vessels according to the image high score regions when compared with predefined vessel classes (Figure 6). YOLOv3-tiny is a real-time detection algorithm developed for devices with a lack of data processing capacity. The model structure is simpler than YOLO, and the detection is very fast. YOLOv3-tiny is a lightweight target detection algorithm applied to embedded platforms.

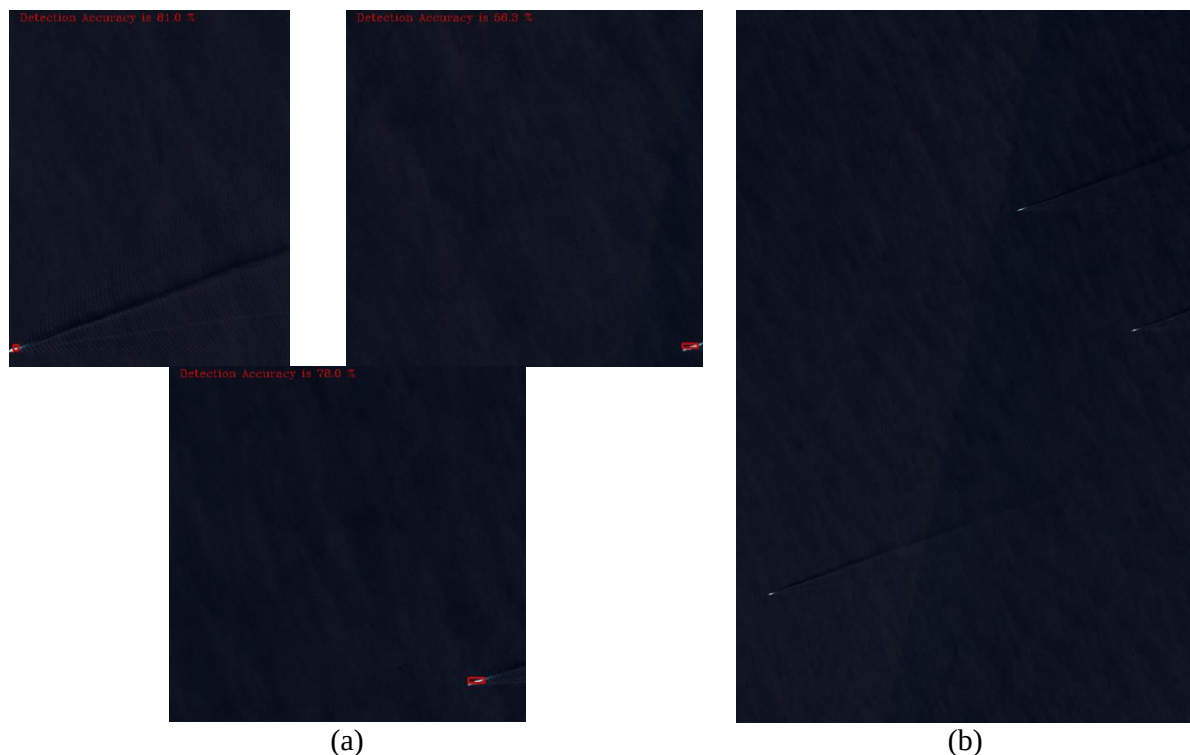


Figure 6: AVA output (a), input image (b)

During the test stage, a single neural network is applied to the whole image. Hence, the model looks for a full image and separates it into smaller tiles. The bounding boxes (tiles) and probabilities are predicted for each location of the image. The bounding boxes are weighted by the predictions, since it allows the model to look at the whole image during tests, leading to informed predictions for the

global context in the image.

YOLO scores each region based on the similarities with predefined classes (from AIS clustering developed classification). To train the model, a dataset of vessel images was created by converging the information of AIS and satellite imagery. Some of the performance metrics of YOLOv3-tiny trained with the vessel dataset are presented in Table 2.

Table 2: Performance values of YOLOv3-tiny trained with vessel dataset for a Intersection over Union (IoU) value of 30%

Metric	Value
Precision (IoU threshold 30%)	0.74
Recall (IoU threshold 30%)	0.87
F1-score (IoU threshold 30%)	0.80

3.3 CLOUD DETECTION

Onboard cloud detection was one of the main goals of the Φ sat-1 experiment and for this second, follow on mission, it has been decided to expand the original concept going into two different directions:

- First to develop a cloud detection app service to be used by other AI applications to offload them from the task of cloud assessment. Given the limited resources onboard, only cloud free (or images with a cloud coverage within the limits specified by the different users) will be moved to the successive processing stages; the service won't manipulate the data (e.g removing cloud scenes) and will also provide ancillary data that the final user can use as input for further processing down the line,
- Second a proper cloud detection demonstration application, based on the service described in the previous point to be used in the initial stages of the mission for validation and fine-tuning purposes.

The optimized U-Net architecture for cloud detection is exploited [6]. To accelerate the inference process offered by this deep learning model, this has been thoroughly benchmarked, and the second convolutional layer was removed, with minor impacts on the detection accuracy (the FPS was increased from 1.62 fps to 5.57 fps here). The models were benchmarked on the Intel Movidius Myriad-2 connected by the USB port to a PC computer (the inference time was measured for a 19.4 x 19.4 km scene, and it amounted to approximately 12 seconds). The segmentation quality of the models was verified over the unseen test data (obtained using a GPU and Myriad), and it was quantified using classic metrics, including the Dice coefficient, accuracy, precision, recall, specificity and the Jaccard's index (elaborated for the Sentinel-2 and simulated Φ sat-2 multispectral imagery).

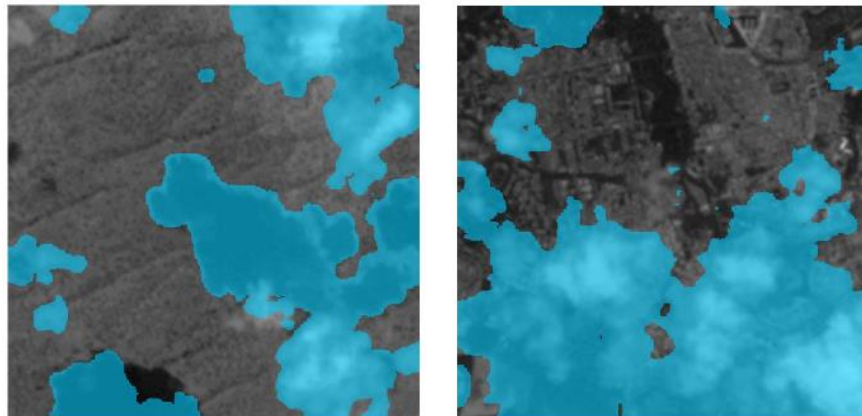


Figure 7: Examples of cloud masks from the processing of the training dataset

The results showed that the optimized U-Net model offers high-quality multi-class cloud segmentation (clouds, semi-transparent clouds, cloud shadows and background, Figure 7) reaching 0.81 categorical accuracy (for the original KappaZeta test set [7]) and fast operation.

3.4 DEEP COMPRESSION

The Deep Compression application performs image compression exploiting a Convolutional AutoEncoder (CAE) i.e., a Convolutional Neural Network (CNN) with an AutoEncoder (AE) structure. The model was developed to demonstrate how onboard AI post processing can reduce the amount of data to be sent to the ground with a limited information loss. The compression of the image is performed on-board: the model accepts as input single band images acquired by the multispectral camera which are compressed through the encoder part of the CAE. Reconstruction is performed on the ground by means of the dedicated decoder.

AEs are Neural Networks (NNs) with a symmetrical structure consisting of an encoding and a decoding part, with one or more hidden layers, and an internal bottleneck layer that forces a compressed knowledge representation of the original input. Essentially, AEs transform input data to a lower dimensional representation and ensure the reconstruction of the input based on these characteristics. AEs are commonly used for feature learning and data compression [8]. The CAE model on which the Deep Compression application is based is summarized in Figure 8. The encoder comprises blocks of convolutional layers (CONV block) and the innermost bottleneck layer which returns the compressed representation of the input images and determines the Compression Ratio (CR). CR is defined as the ratio between the size of the input image in terms of pixels and the number of units in the bottleneck layer. Symmetrically, the decoder part includes transposed convolutional blocks (DE-CONV block) and return back the output with the same dimension as the input. In the encoder, convolutional layers progressively extract deeper feature maps through an increasing number of filters. Then, the output of the last convolutional layer is flattened to return the compressed representation of the image through the bottleneck layer. In the decoder the reverse process is performed, returning the reconstructed image as final output. The final definition of the model was determined considering the requirements imposed by the mission, especially in terms of memory footprint and computing power usage. The dataset used for training and validating the application is based on SIMERA Sense MultiScape100 CIS simulated acquisitions (Figure 8). After the training phase, the encoder and decoder parts are used separately.

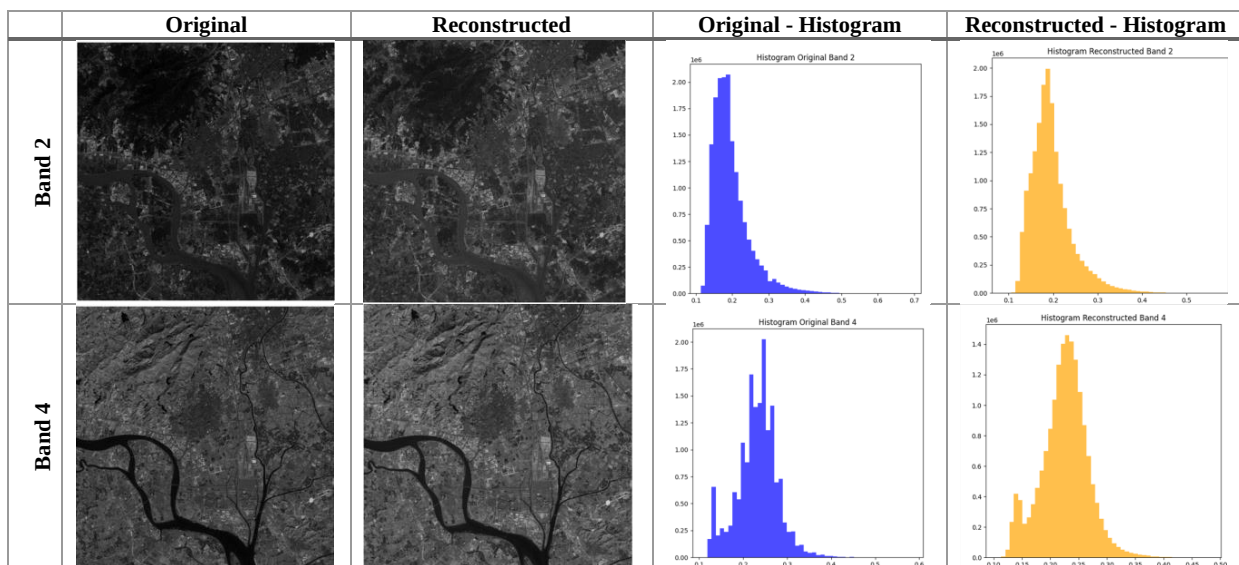


Figure 8: Training of the Deep Compression Application

The quality of the reconstructed image was evaluated using standard metrics, i.e., Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) widely used for lossy compression evaluation. PSNR depends on Mean Squared Error (MSE), and higher PSNR values indicate higher image quality. SSIM is related to the perception of image quality by the human eye and ranges between 0 and 1 [9]. Additionally, an applicative metric was implemented that aims at assessing the practical use of the reconstructed images. The images compressed and reconstructed through the CAE were indeed used in a real application scenario, that is semantic segmentation. More in detail, the decompressed images were used to predict building masks along with the original acquisitions. The obtained masks were then compared to assess the differences between the two cases, confirming that the decompressed images can be used with the same results as the originals as presented in Table 3.

Table 3: Results from the ground testing campaign of the Deep Compression Application

Metrics	Value		
		FlatSat	Local - Myriad
SSIM	Band 0	0.919	0.919
	Band 1	0.917	0.917
	Band 2	0.908	0.908
	Band 3	0.972	0.972
	Band 4	0.964	0.964
	Band 5	0.962	0.962
	Band 6	0.895	0.895
CR		89.6	89.6
Processing time		30.85 s	26.29 s

3.5 Applications selected from the OrbitalAI challenge

The OrbitalAI challenge was run in the course of 2023 and allowed the down-selection of 2 additional applications to complement those already in since the beginning of the project. The first app was created by a team led by the IRT Saint Exupéry Technical Research Institute in France. Its machine learning algorithms scan the satellite's imagery to spot anomalies in marine ecosystems, with trials showing considerable success in characterising oil spills, algal blooms and sediment flooding.

The second application is called PhiFire AI and allows for the detection of wildfires that delivers swift early-warning information to response teams. Created by Thales Alenia Space in Italy, the tool provides a classification report that identifies not only the fire itself but also nearby safe, burnt and water areas.

Both applications are currently being prepared to be uploaded to the spacecraft during the commissioning phase.

4 CONCLUSIONS AND FUTURE WORK

This paper presents the current status of the Φ sat-2 mission, after the successful closure of its QAR. A description of the payload data handling architecture is provided emphasizing the need for an additional processing step to be applied to the data before the AI inference. This step represents one of the major achievements of the mission development because it enables the generation onboard of data that can be directly utilized by the end users just like in the case of the AVA and SAT2MAP applications. The implemented architecture allows the chaining of the various applications as well with the goal of further improving the initial data reduction target throughout Cloud Detection and Deep Compression. The Φ sat-2 spacecraft is currently integrated inside its deployer and it is waiting for being moved to the launch site in early June 2024.

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